



香港大學

THE UNIVERSITY OF HONG KONG

Development of a Compact Vision-Based Tactile Sensor for Robotic End Effectors

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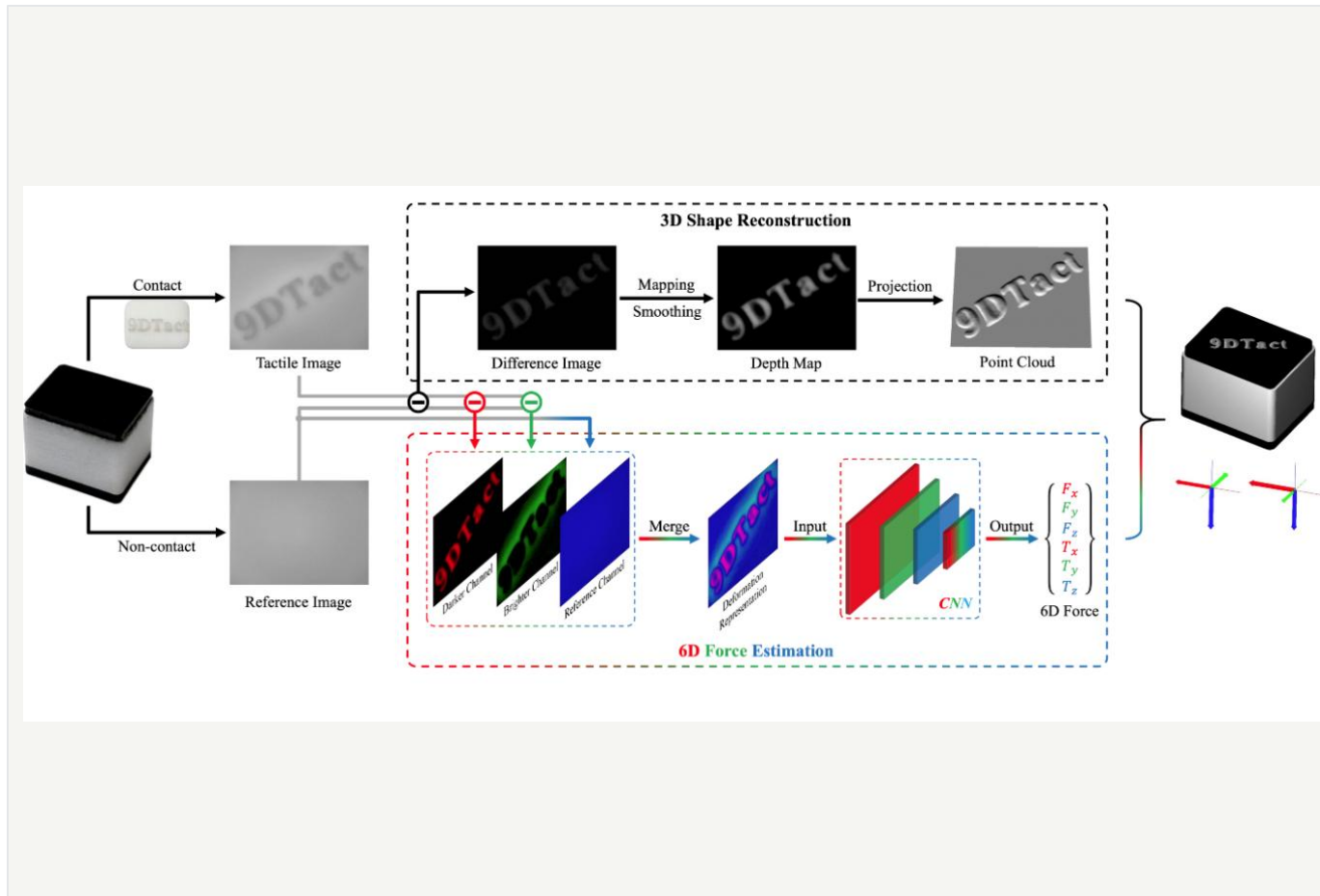
Co-supervisor: Dr. WEI Yan

7 August 2026



Completed compact sensor prototypes

9DTact Sensing Principle



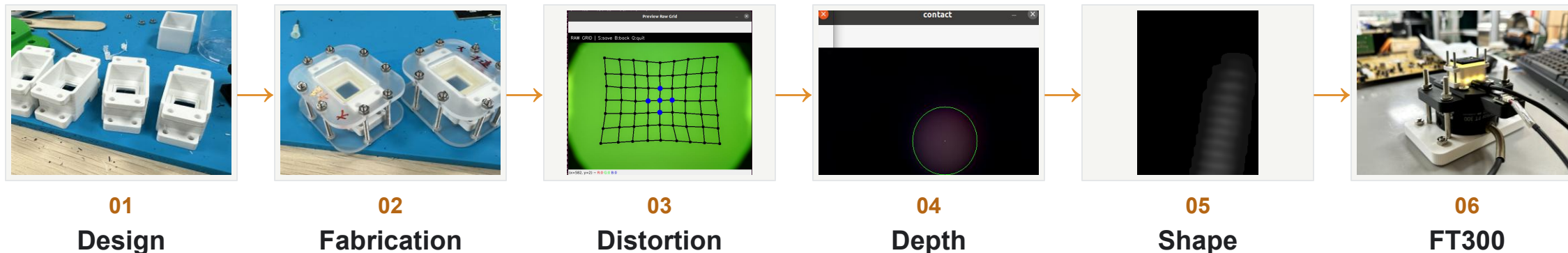
Internal camera observes contact-induced optical change



Internal image keeps contact shape visible

- Reference image records the surface before contact
- Contact image captures the optical change
- Difference image maps to depth and 3D shape
- RGB changes support six-axis force learning

Project Aim and Technical Scope



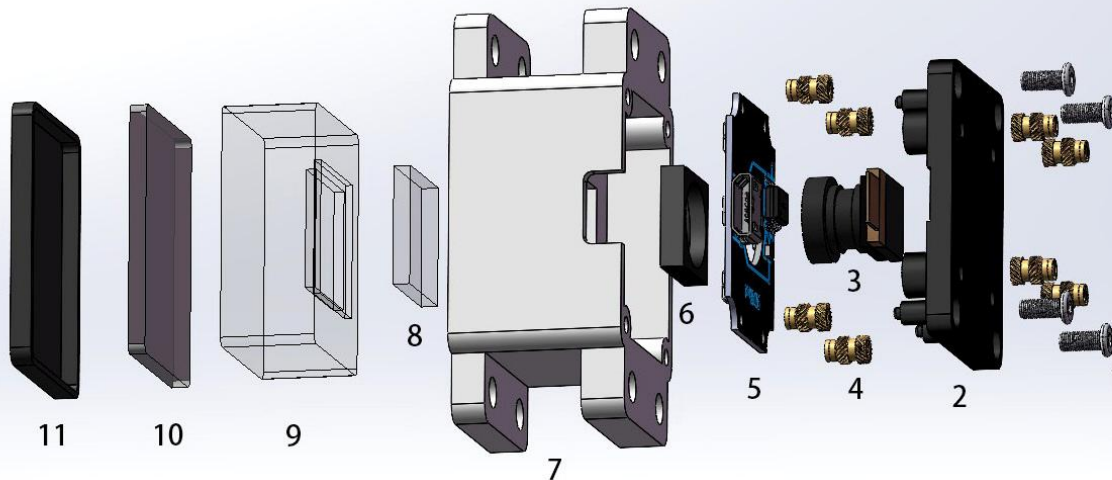
Completed in this project

- Compact sensor design and physical fabrication
- My manual-point distortion script
- My depth script and fine 3D test
- FT300 platform and initial data test

Planned next stage

- Large image–FT300 dataset
- Six-axis model training
- Model evaluation after training

Initial data test completed; no force accuracy claimed.



Exploded view of the complete optical-mechanical stack

Component groups

- G25/G5/G50 contact layers
- Acrylic window supports optics
- Eight-LED PCB surrounds camera
- Camera, ring and sensor base
- Enclosure, mounts and cable outlet

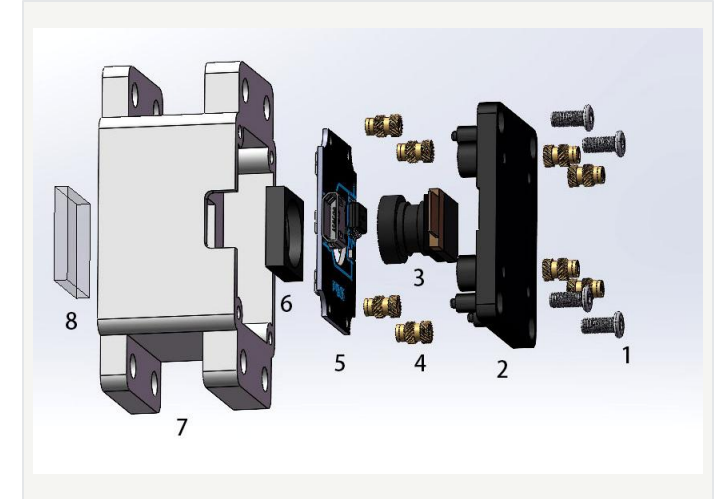
Contact → optical change
Camera image → depth map



Compact camera and separate decoder



Eight-LED PCB around the lens

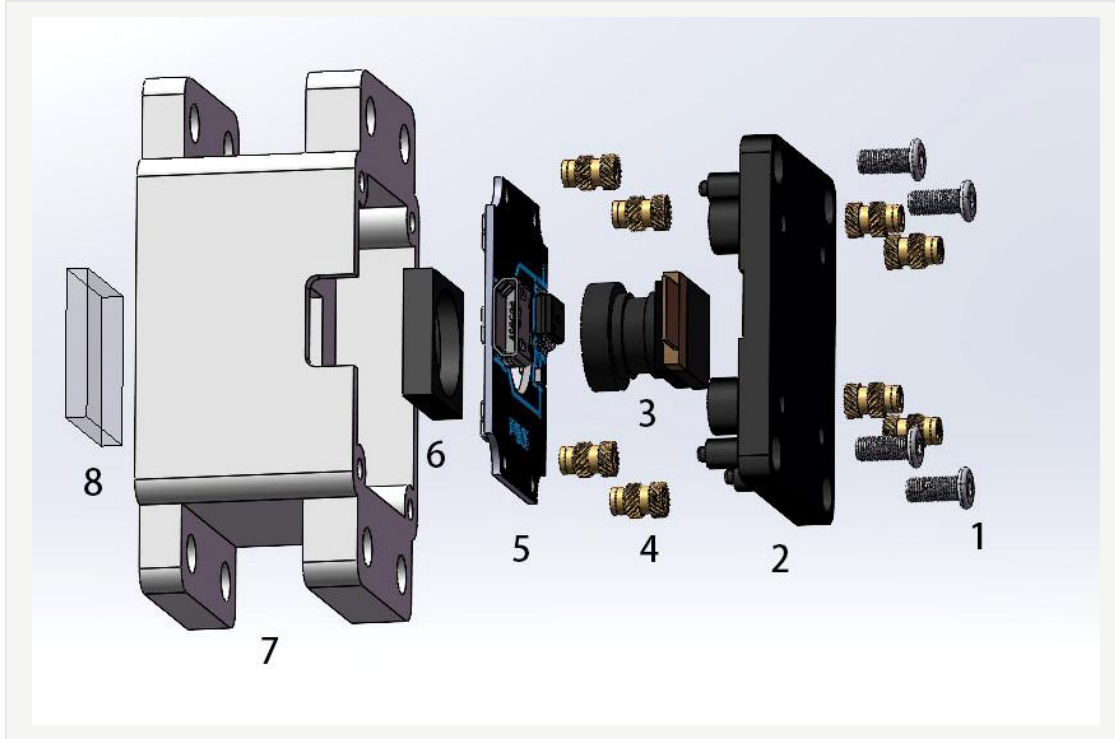


Camera locating and retaining parts

Why the structure and calibration changed together

- Camera head stays inside the sensor
- Eight LEDs surround the central lens
- New optics require sensor-specific calibration
- Decoder board remains outside the housing
- Lens and LED alignment affect every image

Mechanical Housing and Cable Protection



Locating, retaining and mounting structure

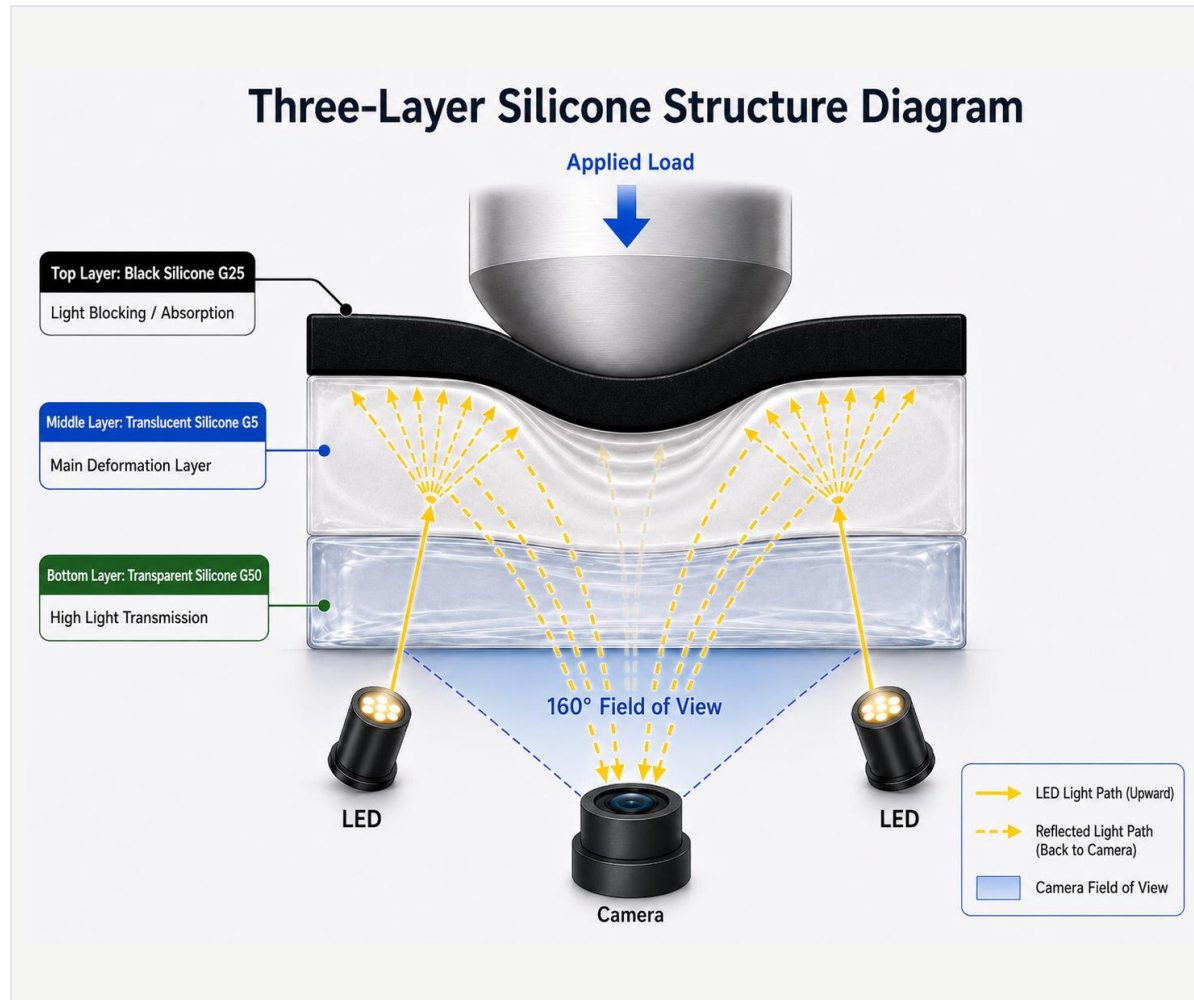
- Housing fixes camera working distance
- Cable clearance prevents screw pinching



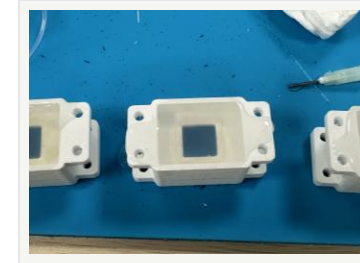
Printed housings prepared for assembly

- Window supports the tactile interface
- Cable stays outside the main load path

Three-Layer Silicone Interface



Each layer has a different optical and mechanical role



G50 - transparent support

Main optical path and support



G5 - soft layer

Concentrates local deformation



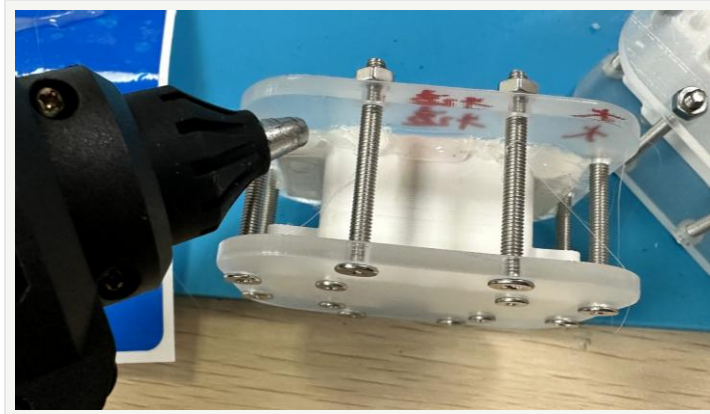
G25 - black contact surface

Blocks ambient light at contact

Mould and Fabrication Process



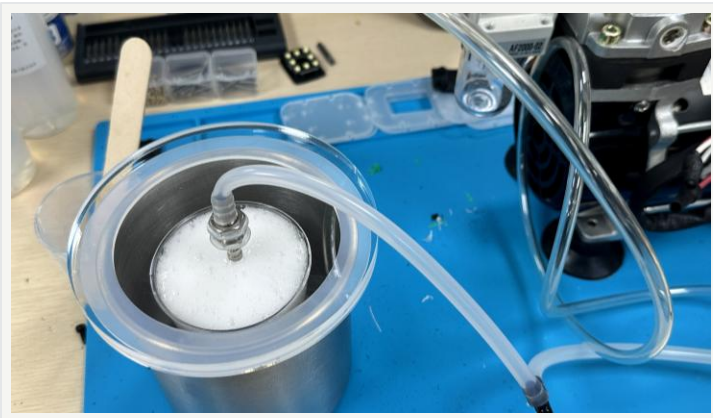
1 Detachable mould



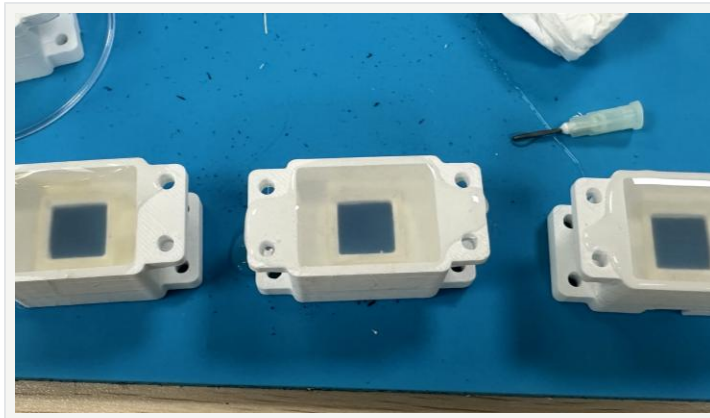
2 Hot-melt seam sealing



3 Silicone mixing



4 Vacuum degassing



5 Layer-by-layer casting



6 Curing, demoulding, assembly

Main controls: leakage · air bubbles · layer separation · demoulding damage

Completed Sensor Prototypes



Black tactile surface · Printed enclosure · Flex cable · External decoder · Mounting lugs

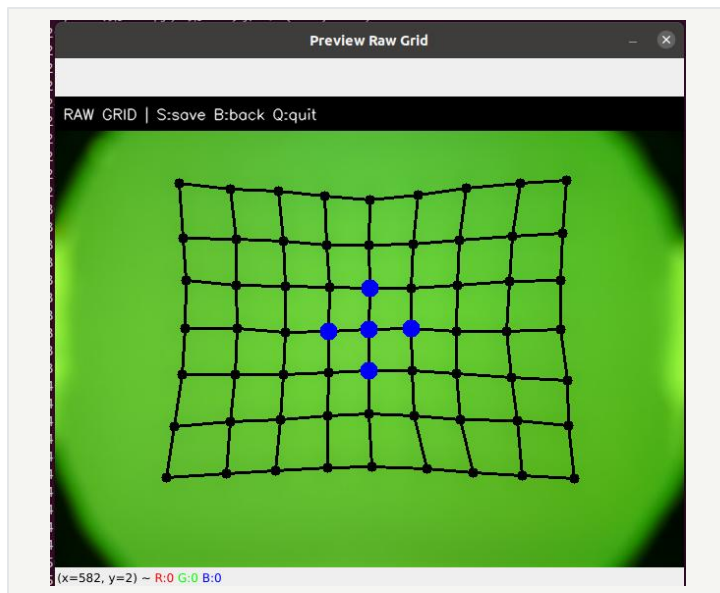
Assembly milestone

- Two complete prototypes assembled
- Camera, LEDs and silicone integrated
- External decoders remain accessible
- Approx. design size: 42 × 38 × 34 mm

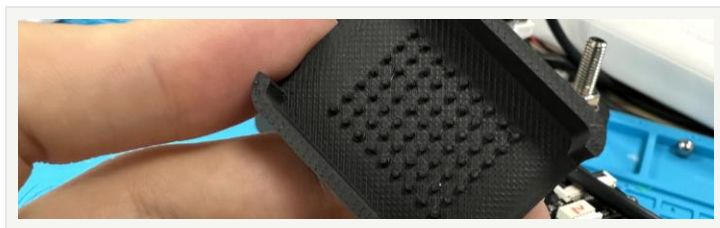
Physical hardware, not a concept rendering

My Camera Distortion-Correction Script

Observed problem

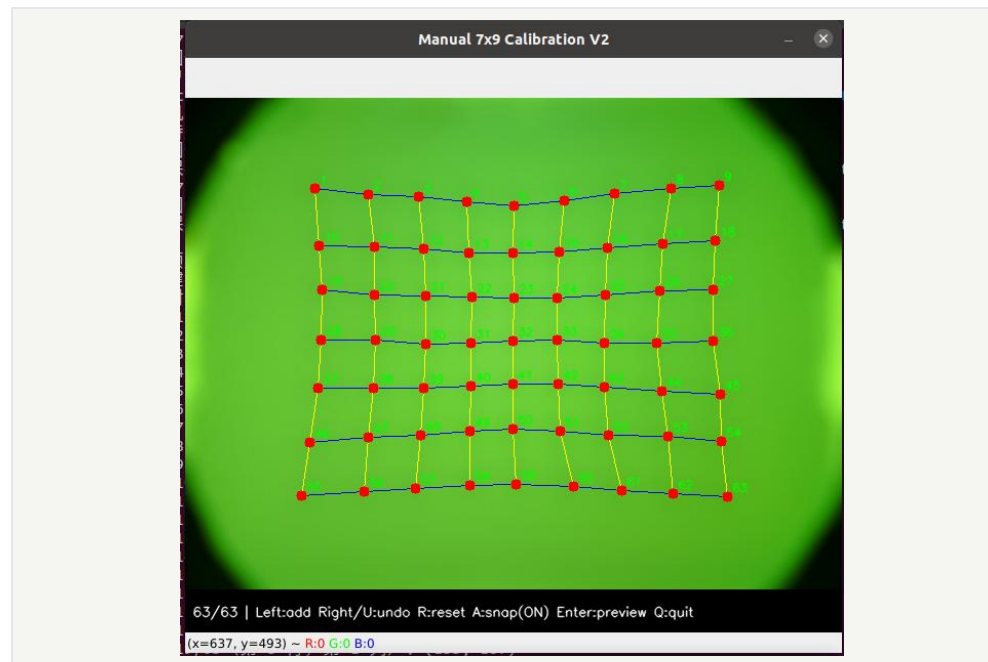


Wide-angle distortion and dark corners

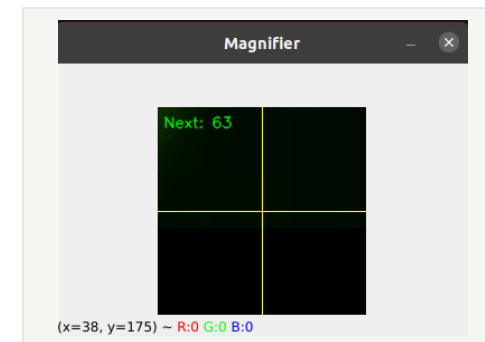


Physical 7 × 9 calibration board

Manual 7 × 9 point selection



63 manual points selected after full assembly

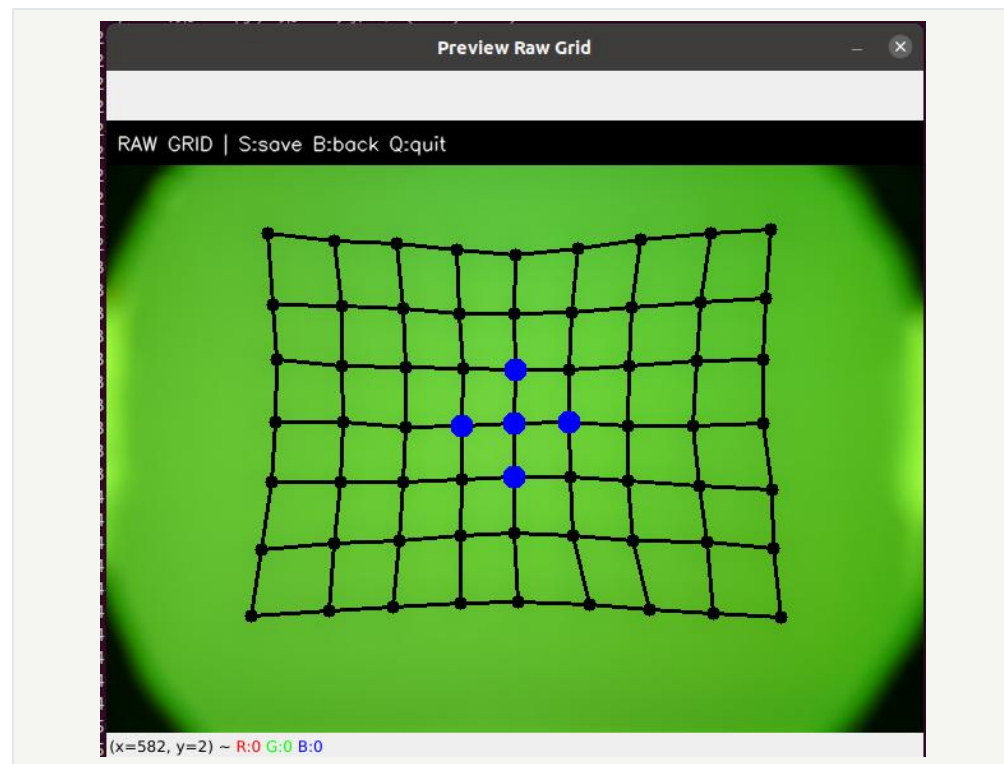


Zoom and order check

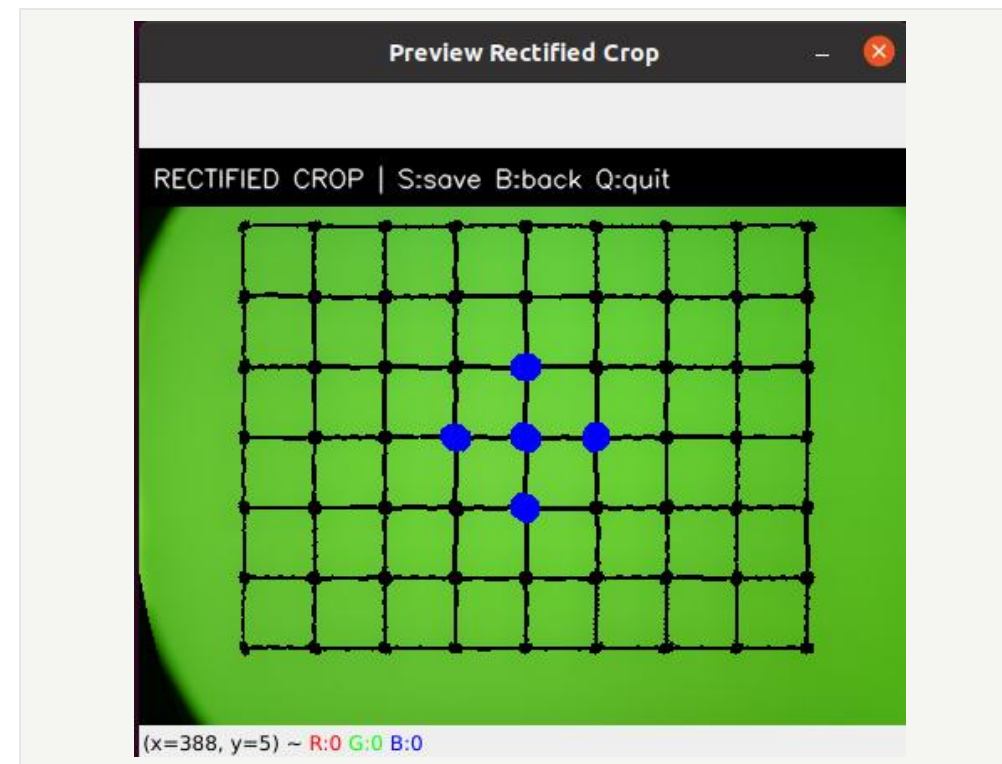
Problem: distortion and vignetting broke auto-detection.
My script: select 63 points by hand; check with zoom and undo.

RBF Correction from 63 Manual Points

Before: distorted grid



After: rectified central region



1 Pair 63 manual and ideal points

3 Run my script on the full image

2 Fit row and column RBF maps

4 Keep the reliable central region

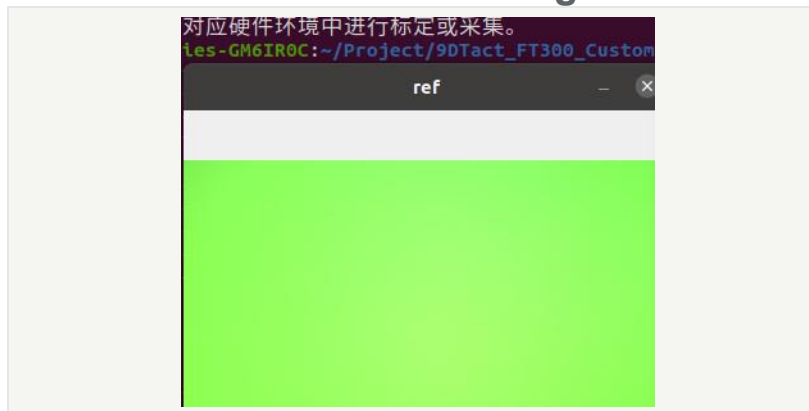
My Depth-Calibration Script

Known reference

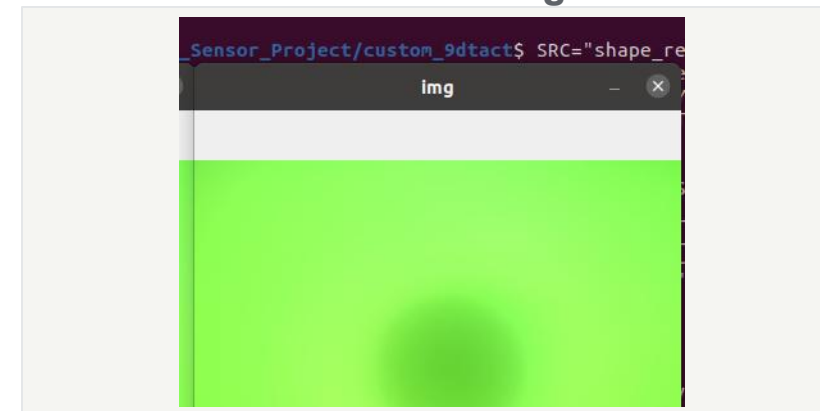


Nominal radius ≈ 4 mm

1 Reference image



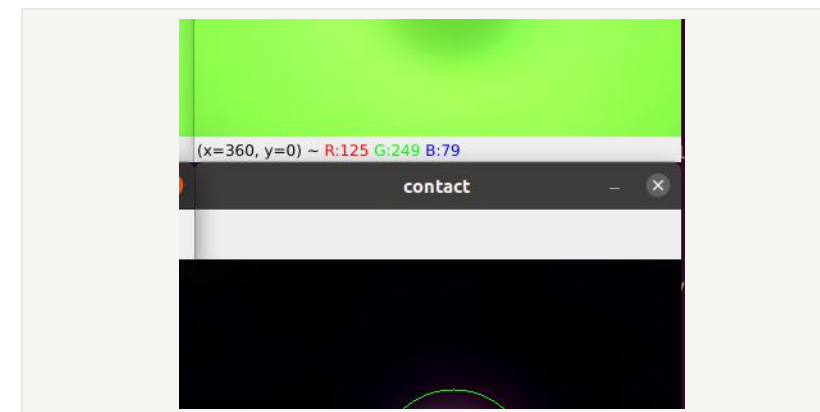
2 Loaded image



3 Image difference



4 Detected contact



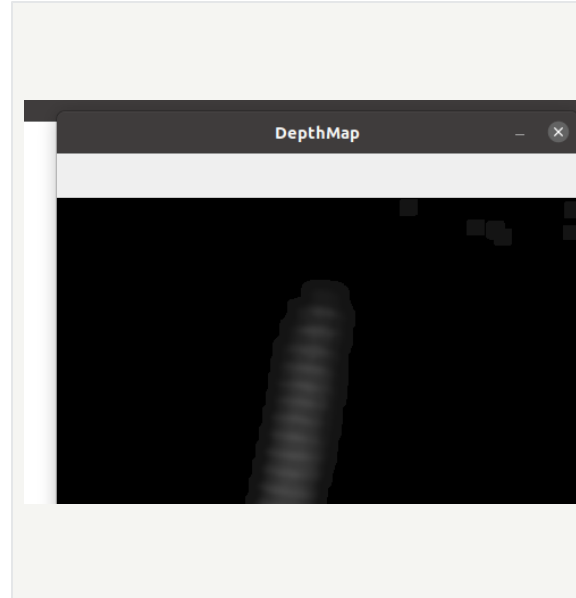
My script: sphere $\rightarrow$ image change $\rightarrow$ depth lookup $\rightarrow$ depth map

Final Physical Test and Fine 3D Reconstruction

Real screw contact



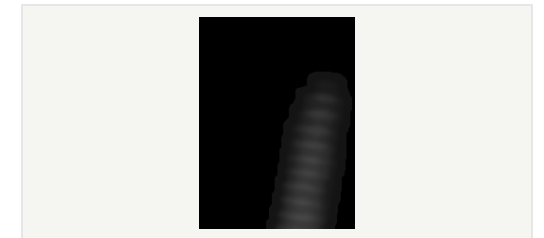
Corrected depth map



Fine 3D reconstruction



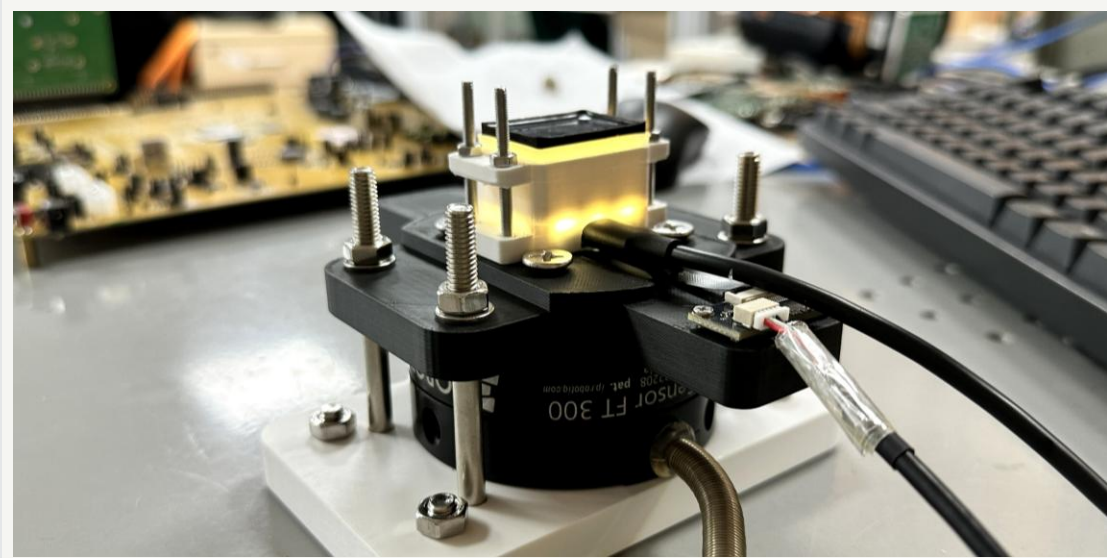
Thread detail



- Final real-object test after image and depth correction
- Fine screw-thread geometry was reconstructed in 3D
- Repeated thread details remained clear

FT300 Data Test and Gripper Integration

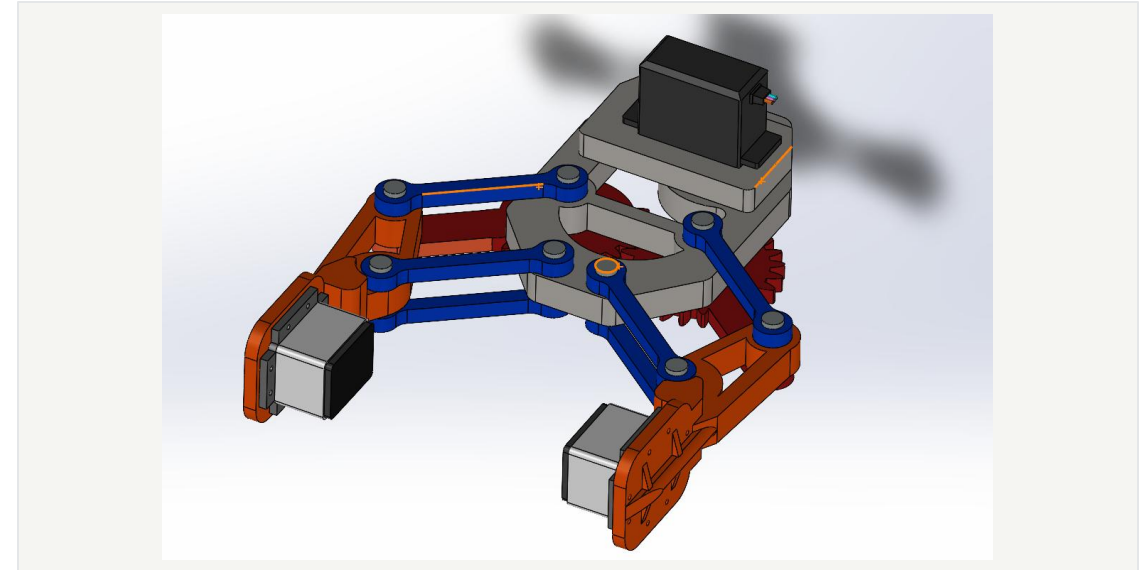
| FT300 data-acquisition platform



Initial data test completed; next, expand the dataset and train the six-axis model.

- Mechanical load path is established
- Cable remains outside the main load path
- Initial FT300 six-axis data test completed

| Two-finger gripper CAD model



- Two sensor models face the grasping region
- CAD model only; no physical grasping test

Conclusions and Remaining Work



Complete engineering path: design → build →
calibrate → validate

Completed

- Two prototypes and the G50/G5/G25 interface
- My distortion script
- Depth script + fine 3D test
- FT300 data test + CAD

Remaining work

- Quantitative depth-error test
- Large paired dataset
- Six-axis model training
- Physical gripper experiments

**A complete engineering implementation
from design to initial validation**



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Thank You.

Questions and Technical Discussion

